

The empirical recurrence for OEIS A239028: recovering a conjecture that is not written down

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Abstract

OEIS A239028 records “Empirical recurrence of order 90”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 109$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 90, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 90 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A239028 is “Number of nX6 0..2 arrays with no element equal to the sum of elements to its left or one plus the sum of elements above it, modulo 3.”. It has offset 1 and begins

1, 7, 22, 189, 757, 7265, 33699, 339315, ...

The entry states:

Empirical recurrence of order 90 (see link above)

As of the “Last modified” line on the live entry (Jun 02 2025 09:23:00, revision 9) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 729$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 109$ of the 729, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 109$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 218$ exact terms of the model returns order 90 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{90} c_i a(n-i),$$

c_1	0	c_{24}	−55518906603418006586705	c_{47}	0
c_2	530	c_{25}	0	c_{48}	−795392897516615705696
c_3	0	c_{26}	1313900069071840304687450	c_{49}	0
c_4	−134480	c_{27}	0	c_{50}	4780667862858765725834
c_5	0	c_{28}	−27457715480135634737700439	c_{51}	0
c_6	21782487	c_{29}	0	c_{52}	−2561087092516997963355
c_7	0	c_{30}	508662960097245520253332068	c_{53}	0
c_8	−2533606480	c_{31}	0	c_{54}	122008139545359539851302
c_9	0	c_{32}	−8379247526134531390533782584	c_{55}	0
c_{10}	225677353189	c_{33}	0	c_{56}	−51539556933556133653094
c_{11}	0	c_{34}	123039840967270285127572228314	c_{57}	0
c_{12}	−16030205615232	c_{35}	0	c_{58}	192383535723183768217463
c_{13}	0	c_{36}	−1613451550808849956760725968363	c_{59}	0
c_{14}	933548483579970	c_{37}	0	c_{60}	−63189961312262468617307
c_{15}	0	c_{38}	18919615815315898054423561327929	c_{61}	0
c_{16}	−45478073895685232	c_{39}	0	c_{62}	181717319036956022304754
c_{17}	0	c_{40}	−198556298011394777921131469325501	c_{63}	0
c_{18}	1881470688703445384	c_{41}	0	c_{64}	−454774102963767009987708
c_{19}	0	c_{42}	1865706748348704807073643549747297	c_{65}	0
c_{20}	−66880532498480134764	c_{43}	0	c_{66}	9833413348280014698552278
c_{21}	0	c_{44}	−15695566617806181756774331379431981	c_{67}	0
c_{22}	2061643663692713131333	c_{45}	0	c_{68}	−182107488327789403480135
c_{23}	0	c_{46}	118163971625695847950731082296444938	c_{69}	0

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 90, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $90 + 4$ terms removed returns order 90 as well, so $\deg q_{\min} = 90 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 21 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once

more on the sequence with its first $90 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture’s statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 90 that this sequence satisfies — and that family turns out to have exactly one member.

References

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